Humanoid Robots in Healthcare: Current Systems, Research Frontiers, and Clinical Opportunities

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Humanoid robots capable of bimanual manipulation and natural language instruction are commercially available, yet no full-size bipedal humanoid holds FDA clearance or CE marking for any clinical application. This survey covers the 2022–2026 humanoid healthcare literature: 14 qualifying papers identified, 25 platforms assessed, regulatory pathways analyzed, and ISO 16290:2013 Technology Readiness Level (TRL) criteria assigned to five clinical domains. All domains are at TRL 3–4. The three roles nearest TRL 5 are teleoperated clinical assistant, rehabilitation demonstrator, and eldercare companion. The binding barrier is the lack of accepted bipedal gait safety standards for patient-proximate environments.

CCS Concepts: • **Computer systems organization** → **Robotics**; • **Computing methodologies** → *Artificial intelligence*; • **Social and professional topics** → *Medical informatics.*

Additional Key Words and Phrases: humanoid robots, healthcare robotics, embodied AI, large language models, human-robot interaction, clinical deployment, medical robots, technology readiness level, regulatory pathway

ACM Reference Format:
Anonymous Author(s). 2026. Humanoid Robots in Healthcare: Current Systems, Research Frontiers, and Clinical Opportunities. *ACM Comput. Surv.* 0, 0, Article 0 (2026), 32 pages.

1 Introduction

Full-size bipedal humanoids can now execute clinical procedures in the laboratory. A Unitree G1 teleoperated by a non-clinician completed seven distinct procedures: auscultation, ultrasound, intravenous catheter insertion, and four others [4]; a second system performed laparoscopic surgery on a cadaveric torso [49]; a third acted as a surgical first assistant in a cadaveric bilateral sphenoidectomy [21]. Not one of these systems holds U.S. Food and Drug Administration (FDA) clearance or Conformité Européenne (CE) marking for any clinical application. No registered clinical trial has enrolled a patient for a study involving a full-size bipedal humanoid platform. Two things converged after 2022 that are beginning to close this gap.

The first is hardware: 13 commercial platforms were released between 2022 and 2026, crossing the payload, dexterity, and locomotion threshold required to attempt clinical tasks in human-scale environments. The second is software: large language models (LLMs) and vision-language-action (VLA) models removed the per-task programming requirement that had previously made general-purpose humanoid deployment impractical. Instruction-following, task generalization, and adaptive response to unstructured environments became achievable without months-long engineering cycles. The result is a field that moved, in roughly three years, from proof-of-concept locomotion to laboratory execution of clinical procedure sequences. Together these make clinical deployment a question of *when and how*, not *whether*.

No existing review covers this territory. Cunha et al. [23] is the closest: a 2025 scoping review of 27 studies with a November 2023 search cutoff, which predates the LLM/VLA integration wave entirely. Cao [16] and Liu [52] do not distinguish bipedal humanoids from wheeled social robots or surgical arms, so their clinical gap analysis applies to a different class of systems. Earlier healthcare robot reviews [6, 35, 60, 64, 70] treat platform types interchangeably,

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2026. Manuscript submitted to ACM

obscuring the form-factor-specific barriers that determine clinical deployability. The result is that the 2022–2026 humanoid healthcare literature has not been synthesized.

This review covers bipedal, free-standing humanoid robots of approximately 150–200 cm height designed for general-purpose manipulation and locomotion in human-scale environments, with literature from January 2022 to March 2026. Wheeled social robots (Pepper, ARI), sub-100 cm platforms (NAO), surgical arms (da Vinci), and powered exoskeletons are excluded; each exclusion is argued on form-factor grounds in Section 2. RHP Friends [7] is the only non-commercial academic platform retained without caveat. The systematic search ran across five databases: arXiv (cs.RO, cs.AI, eess.SY), PubMed, IEEE Xplore, ACM Digital Library, and Google Scholar. The primary query combined platform and application terms: (*humanoid robot* OR *bipedal robot*) AND (*healthcare* OR *medical* OR *clinical* OR *nursing* OR *rehabilitation* OR *surgery*), supplemented by domain-specific secondary searches and citation-list hand-search. Title and abstract screening of 147 candidate papers yielded 52 full-text reviews; applying inclusion and exclusion criteria produced 14 qualifying papers as the primary evidence corpus; the full survey draws on 100 references covering background literature, technical foundations, and regulatory sources. The sparsity of this primary corpus is itself a central finding: the evidence base for bipedal humanoid clinical research is narrow and concentrated. Concurrent with the literature search, a platform survey identified 13 commercial and 12 academic systems meeting scope criteria, with specifications verified against manufacturer documentation or peer-reviewed publications.

This review makes four contributions. First, we provide the first systematic review of humanoid robots in healthcare covering the 2022–2026 LLM/VLA era, with Technology Readiness Level (TRL) assignments per ISO 16290:2013. Second, we assess the capabilities of 25 bipedal humanoid platforms (13 commercial, 12 academic) against five clinical domain requirements. Third, we analyze regulatory pathways covering FDA De Novo, EU Medical Device Regulation (MDR) 2017/745, and EU AI Act 2024/1689. Fourth, we introduce the Humanoid Clinical Deployment Readiness (HCDR) framework, which links technical maturity to regulatory pathway for five clinical roles. The remainder of this paper is organized as follows: Section 2 provides background; Section 3 surveys current humanoid capabilities; Section 4 analyzes what healthcare demands from humanoid robots; Section 5 examines clinical deployment challenges; Section 6 discusses future directions; Section 7 concludes.

2 Background

Before assessing the clinical evidence, three core developments must be understood: the healthcare workforce crisis that creates demand for robotic assistance, the humanoid form-factor argument that makes bipedal platforms specifically relevant, and the LLM/VLA inflection that makes general-purpose clinical deployment conceivable for the first time. Two supplementary elements complete the context: a historical trajectory situating current platforms within the arc of bipedal robotics, and a structured comparison of humanoid against non-humanoid robotic alternatives.

2.1 The Healthcare Imperative

The global healthcare workforce is under structural strain that technology alone cannot resolve, but that technology may help mitigate. The World Health Organization projects a shortage of approximately 10 million healthcare workers by 2030, concentrated primarily in low- and middle-income countries but with significant pressure also on high-income systems where demographic aging compounds absolute workforce insufficiency [95]. The root cause is well understood: populations in Organisation for Economic Co-operation and Development (OECD) countries are aging faster than working-age cohorts are growing [69]. In Japan (the most extreme case), persons aged 65 and over already represent approximately 30% of the total population, a proportion that continues to rise under below-replacement fertility rates [13]. The METI 2023 robotics roadmap explicitly identifies nursing-capable robots as a national strategic target by 2030, reflecting governmental acknowledgment that the eldercare workforce gap is structurally unsolvable by recruitment alone.

Nursing workforce shortages present a compounding challenge [48]. The care gap operates at two intersecting levels.

First, the absolute number of care hours demanded rises with the prevalence of chronic disease, multimorbidity, and frailty in an older population: an individual with dementia and comorbid heart failure requires qualitatively more nursing time than a healthy adult, and the number of such individuals is growing rapidly. Second, the tasks that constitute professional care (physical repositioning of patients, vital sign monitoring, medication administration, rehabilitation exercise guidance, and social engagement) are labor-intensive and resist substitution by currently deployed information technology. Electronic health record systems and remote monitoring platforms reduce administrative overhead but do not reduce the need for physical presence at the bedside.

Automation is increasingly discussed as a necessary complement, not a replacement, for human care. The framing matters: the goal is not to displace nurses or physicians, but to extend their effective reach by delegating tasks that are physically demanding, repetitive, or logistically routine, freeing trained clinicians for higher-order judgment and human connection. Wheeled mobile robots already handle medication transport and specimen delivery in hundreds of hospitals. Exoskeletons assist rehabilitation and reduce caregiver musculoskeletal strain. Social robots provide cognitive stimulation and companionship in dementia care settings. Each of these automation layers is real and documented. What remains largely unrealized is a robot capable of performing the physical care tasks (patient repositioning, standing assist, equipment handling, clinical examination) that currently require a human body in proximity to the patient. This is where the humanoid form factor becomes relevant. Socially assistive robotics [25, 76] has addressed parts of this gap through social interaction and coaching, but not the physical care layer; the humanoid form factor is the candidate solution specifically for physical assistance.

2.2 Why the Humanoid Form Factor

For the purposes of this survey, a *humanoid robot* is defined as a bipedal, free-standing robot with a human-proportioned morphology: head, torso, two arms, and two legs, typically ranging from 150 to 200 cm in height. This definition explicitly excludes social robots such as Pepper (120 cm, wheeled base) and NAO (58 cm, child-scale), surgical manipulator arms such as the da Vinci system, and wearable exoskeletons. The rationale for this scope restriction is not hierarchical (social robots and surgical robots are clinically important and are discussed in Section 2.5), but conceptual: the form-factor hypothesis is specifically about whether human-scale bipedal morphology confers advantages in human-built environments that smaller or non-mobile systems cannot provide.

Three distinct lines of argument support the humanoid form factor's suitability for healthcare deployment.

Environmental compatibility. Hospitals, care homes, and residences are built for human-sized bipedal operators. Doorways, corridors, stairways, beds, examination tables, supply cabinets, and clinical instruments are all dimensioned for a person approximately 160–180 cm tall with two-handed reach. A bipedal humanoid can, in principle, navigate these environments without infrastructure modification: walk through standard doorways, climb stairs, reach across a standard hospital bed, and operate existing clinical tools (stethoscope, bag-valve mask, otoscope, ultrasound probe) designed for human hands. The technical demonstration by Atar et al. [4] explicitly grounds this observation: the Unitree G1 demonstrated seven medical procedures using existing clinical instruments, without facility modification.

Neuroscientific basis for human-robot interaction. Action observation therapy (AOT), a rehabilitation technique in which patients observe biological motion to activate sensorimotor circuits, rests on the mirror neuron hypothesis:

observing purposive action activates similar neural circuits as performing that action [77]. Nguyen et al. [65] provide the most direct empirical evidence available: EEG recordings from healthy participants during observation of a humanoid robot performing reaching and grasping actions showed sensorimotor response patterns comparable to those elicited by observing a human agent perform the same actions. This finding provides neurophysiological grounding for the hypothesis that the humanoid form specifically may be required for AOT efficacy. A complementary behavioral finding comes from Sayis and Gunes [80], who demonstrated that only the humanoid robot condition (not a voice-only assistant) produced improvements in mood and increased self-disclosure in a therapeutic writing study (n = 42), attributing the difference to the physical embodiment of the humanoid.

User acceptance and eye-level interaction. Patients in clinical settings are frequently in positions of physical vulnerability. The SPRING project [2] (*scope caveat*: ARI is a wheeled platform, not bipedal) found that users in a real geriatric day-care facility explicitly cited upright, eye-level conversational presence as important for dignified interaction, a form-factor finding directly relevant to bipedal humanoid design regardless of the mobility substrate. Additional contextual evidence comes from research on the HRP-4C platform, which demonstrated that gait naturalness and facial expression generation systematically affected observer ratings of approachability and social presence [16].

These findings collectively support, but do not yet conclusively prove, the form-factor hypothesis at clinical scale. The evidence is encouraging; it is not sufficient to substitute for clinical validation.

2.3 Why Now: The LLM/VLA Inflection Point

Prior to approximately 2022, humanoid robots required explicit, hand-crafted programming for every behavioral task. A robot capable of performing a stethoscope placement procedure had to be programmed, step by step, for that specific procedure. Deploying the same robot for a second procedure required a substantially new programming effort. This approach fundamentally limited humanoid deployment in clinical environments, which are intrinsically unstructured.

The emergence of large language models (LLMs), vision-language models (VLMs), and vision-language-action (VLA) models has changed this constraint. GPT-4 and its successors demonstrated that natural language can serve as a general-purpose task specification interface, capable of interpreting novel instructions and decomposing them into action plans [68]. Physical Intelligence's π_0 and $\pi_{0.5}$ models introduced flow-based VLA architectures mapping language instructions and visual observations directly to robot actions, trained on thousands of hours of real robot demonstrations [9]. Google DeepMind's Gemini Robotics applied multimodal foundation models to general-purpose manipulation with dexterity approaching human-level performance on standardized tasks [28]. Universal Manipulation Interface (UMI) demonstrated that physically deployed robot teaching without robot-specific teleoperation hardware is viable for generalizable manipulation [20]. OpenVLA [44] demonstrated that open-source, community-accessible VLA models could achieve meaningful generalization from limited task demonstrations. 3D-VLA [101] extended this to three-dimensional grounding for manipulation in complex environments. HiRT [94] demonstrated hierarchical transformers for whole-body robot control with improved task generalization. RT-2 [11] established that vision-language foundation models trained on internet-scale data transfer task-relevant knowledge to robotic control without task-specific fine-tuning. Embodied AI surveys covering the healthcare vertical confirm that this software inflection is accelerating clinical relevance [51, 52, 58].

Yuan et al. [98] demonstrated this integration pattern in a dementia care context, combining reinforcement learning with a large language model to generate adaptive, personalized robot behavior. Although Pepper is a wheeled robot (not within our bipedal scope), the LLM integration framework, specifically its ability to model patient-specific cognitive-emotional states and adapt behavior accordingly, is directly applicable to bipedal humanoid platforms.

The commercial humanoid wave of 2022–2026 coincides with this LLM inflection, and the coincidence is not accidental. Twelve of the thirteen commercial humanoid systems surveyed in Section 3 have confirmed or announced LLM/VLA integration as of 2026. The design assumption of current commercial humanoids is language-conditioned, generalist behavior: these systems are not being built as specialized single-task machines, a prerequisite for clinical task diversity.

2.4 Historical Evolution

The history of humanoid robotics over the past three decades can be organized into three eras, followed by the nascent healthcare entry period this review covers.

Locomotion proof-of-concept era (1996–2010). Honda's ASIMO (2000) demonstrated that dynamic bipedal walking was achievable with then-current actuation and control technology [16, 79]. The AIST HRP series (HRP-2 through HRP-5P, 1998–2019) established the research infrastructure for whole-body motion planning [43]. Preview control of the zero-moment point (ZMP) provided the mathematical foundation for stable bipedal gait generation [42]. These systems proved that bipedal locomotion was a solvable engineering problem, with no meaningful manipulation capability or path to clinical relevance. ASIMO was retired in 2022.

Manipulation and integration era (2010–2020). The DARPA Robotics Challenge (DRC, 2012–2015) revealed the gap between locomotion capability and reliable task execution. Boston Dynamics' hydraulic Atlas became the reference platform for whole-body manipulation research and produced dynamic capability demonstrations while revealing that force-controlled manipulation at clinical precision remained largely unsolved [16]. The decade consolidated learning-based locomotion control, particularly deep reinforcement learning policies trained in simulation and transferred to hardware. Recent work demonstrates that transformer-based policies trained on motion-capture data enable zero-shot locomotion on commercial humanoids [74]. Research platforms such as TALOS [88] and iCub3 further expanded the manipulation research base during this period. Whole-body expressive motion control on commercial humanoids was demonstrated with high fidelity by 2024, enabling natural interaction behaviors relevant to care settings [18].

Commercial scale-up era (2022–present). The confluence of electric actuator maturation, learning-based locomotion policies, and LLM integration produced the current commercial humanoid wave. Thirteen distinct commercial humanoid systems reached hardware maturity between 2022 and 2026. Global investment in humanoid robotics grew rapidly through 2024, driven by the convergence of electric actuator maturation and LLM integration, with confirmed commercial deployments concentrated in automotive and logistics applications. As of March 2026, none of these thirteen systems has been deployed in a confirmed clinical patient care context.

Healthcare entry point (2024–2026). Atar et al. [4] demonstrated bimanual teleoperation of a Unitree G1 across seven clinical procedures with approximately 70% aggregate success, the first systematic evaluation of a current-generation commercial humanoid against a clinically meaningful task battery. Cho et al. [21] extended this to a cadaveric endoscopic sinus surgery at Johns Hopkins University, the first documented use of a modern commercial bipedal humanoid in an actual (cadaveric) surgical procedure. This survey covers this nascent period.

Figure 1 situates these milestones in the broader thirty-year arc of humanoid robot development.

2.5 Humanoid Robots vs. Other Medical Robots

A principled comparison with existing clinical robot categories is necessary to locate the humanoid value proposition precisely.

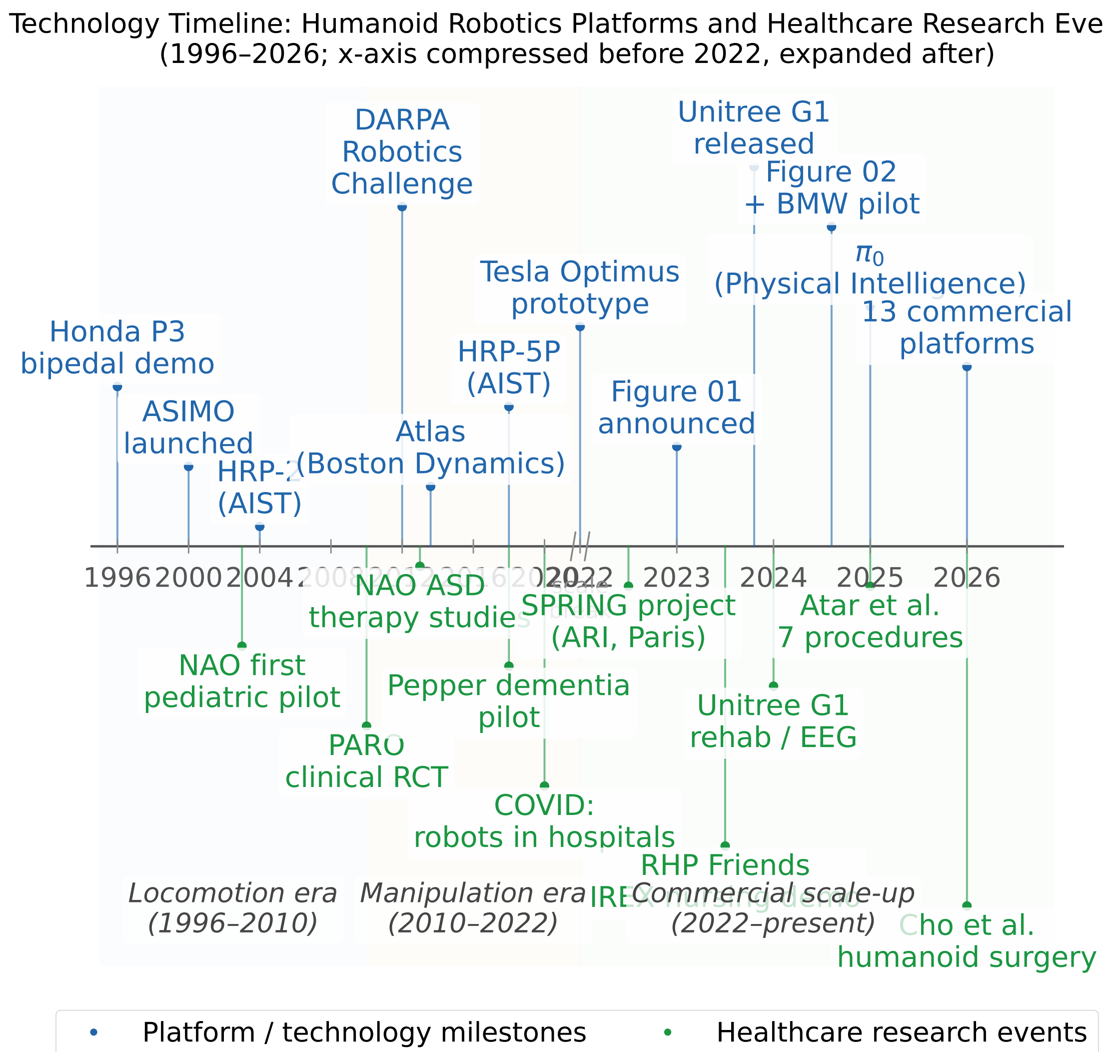

Fig. 1. Technology timeline: humanoid robotics platform milestones (upper track, blue) and healthcare research events (lower track, green), 1996–2026. Shaded regions demarcate three developmental eras: locomotion proof-of-concept (1996–2010), manipulation and integration (2010–2022), and commercial scale-up (2022–present). Healthcare research events lag platform capability by approximately two years in each era.

2.5.1 Social Robots: NAO, Pepper, PARO. Social robots [24, 27] have the deepest clinical evidence base in robotics. NAO (SoftBank/Aldebaran, 58 cm, 25 degrees of freedom (DOF)) has accumulated over 1,000 peer-reviewed publications, with clinical use spanning autism spectrum disorder therapy, elderly care, and hospital patient reception [12]. Pepper has been deployed in approximately 2,000 Japanese healthcare facilities for dementia cognitive stimulation and companion interaction, with embodied conversational agent frameworks increasingly integrated for clinical applications [23, 73, 87]. Observational studies have documented psychosocial benefits of the PARO therapeutic seal robot in elderly residents [93], and an integrative review has identified potential for social connectedness in dementia care [62]. Long-term deployment in care homes has been studied for social connectedness and dementia care [63]. These platforms set the deployment high-water mark for robots in clinical care settings [1]. However, their form-factor constraints are real: NAO at 58 cm cannot reach adult-height work surfaces or perform physical care tasks for adults; Pepper on a wheeled base cannot navigate stairways or use standard clinical beds.

2.5.2 Surgical Robots: da Vinci. The da Vinci Surgical System is FDA-cleared (Class II, 510(k)) with over 10 million procedures completed across urology, gynecology, and general surgery [47, 85]. Robotic surgery adoption has grown substantially over the past decade, though adverse event analysis reveals the importance of systematic safety monitoring even for established platforms [3]. The historical trajectory from laboratory prototype to clinical instrument provides context for humanoid deployment [15]. Surgical robots are not competitors to humanoid clinical assistants; they are complementary systems addressing a fundamentally different need [60]. Da Vinci is stationary, surgeon-controlled, and single-purpose. A humanoid clinical assistant operating in a post-surgical ward addresses the nursing and logistics tasks that da Vinci cannot.

2.5.3 Exoskeletons: ReWalk, HAL, Ekso Bionics. Wearable robotic exoskeletons are FDA-cleared (Class II) for spinal cord injury rehabilitation, with Cochrane-level evidence supporting electromechanical-assisted training for walking recovery after stroke [57]. These devices are established tools in rehabilitation [87]. Exoskeletons assist the patient's own movement and are not substitutable with a humanoid assistant: they are worn by the patient, require near-ambulatory capacity, and cannot lift or reposition a non-ambulatory patient. The GentleHumanoid framework [54] demonstrates the humanoid's complementary role: learning upper-body compliance for sit-to-stand assistance addresses patients for whom exoskeletons are not an option.

2.5.4 Wheeled Mobile Robots: TUG, Moxi. Autonomous wheeled mobile robots address hospital logistics with confirmed operational deployments at major US health systems [35, 60]. These systems are purpose-built for point-to-point transport on flat pre-mapped surfaces. They cannot navigate stairs, interact with clinical equipment not specifically engineered for their interfaces, or perform physical patient care tasks. The humanoid argument relative to wheeled logistics robots is a division of labor argument: wheeled robots solve the logistics layer effectively; humanoids are needed for the physical assistance and care layer.

The four alternative categories collectively frame a finding that motivates this review: no existing clinical robot category combines (a) general ambulatory mobility in human-built environments, (b) bimanual dexterous manipulation of existing clinical tools, and (c) natural language-mediated task specification enabling adaptive behavior in unstructured settings. This combination is the humanoid value proposition for healthcare. The evidence reviewed in this section supports the form-factor hypothesis without constituting clinical proof. The remainder of this survey characterizes where the field stands, what has been demonstrated, and what must be resolved before that promise can be realized.

3 What Can Humanoid Robots Do Today?

The global humanoid robot field in 2024–2026 comprises two largely distinct populations: commercial platforms designed for industrial labor and a set of academic research systems whose provenance spans two decades of bipedal locomotion research. The central empirical finding of this section is that no confirmed clinical deployment of any full-size bipedal humanoid robot exists as of March 2026 [4, 21, 23].

Table 1. Humanoid Platform Capability Comparison (March 2026). N/A = not publicly disclosed. †Research prototype only, not clinical deployment. *Not bipedal under this paper's scope definition; included as deployment benchmark.

Robot	Type	Ht	Total DOF	Payload	LLM Integration	Open SW	Medical Deployment
Figure 02 [26]	Commercial	168	N/A	20 kg	Yes (in-house VLA)	No	None
Tesla Optimus G2	Commercial	173	28+	20 kg	Yes (FSD-derived)	No	None
BD Atlas (elec.)	Commercial	~165	28	30 kg	Yes (Gemini, announced)	No	None
Agility Digit v4	Commercial	175	N/A	16 kg	Announced	Partial	None
Unitree G1	Commercial	132	23–43	N/A	Yes (VLA, open)	Yes	Research proto. [4, 21]†
Unitree H1	Commercial	180	19	N/A	Yes (open)	Yes	None
1X NEO Beta	Commercial	165	75	25 kg	Yes (proprietary)	No	None
Apptronik Apollo	Commercial	173	N/A	25 kg	Yes (Google AI)	No	None
Sanctuary Ph. G8	Commercial	170	N/A	25 kg	Yes (Carbon/Microsoft)	No	None
Fourier GR-2	Commercial	175	53	N/A	Announced	Partial	None
UBTECH Walker X	Commercial	130–145	41	10 kg	Yes (multimodal LLM)	No	None
AgiBot A2	Commercial	175	49+	N/A	Yes (proprietary)	No	None
NEURA 4NE-1	Commercial	~175	N/A	15–20 kg	Yes (cognitive AI)	No	None
RHP Friends	Academic	168	30	N/A	No	Partial	IREX 2023 nursing demo [7]
NAO*	Academic	58	25	N/A	No	Partial	Human-Robot Interaction (HRI)/ASD trials (social only)
Pepper*	Academic	120	N/A	N/A	No	Partial	2,000+ Japan care facilities

Table 1 provides the full specification comparison for all thirteen commercial and selected academic platforms discussed in this section. The analysis below synthesizes capability patterns and clinical relevance across the field rather than characterizing individual platforms.

3.1 Manipulation Maturity

Dexterous manipulation is the primary functional gap between current humanoid capability and clinical task requirements. Across the thirteen commercial platforms surveyed, total articulation ranges from approximately 3 DOF per arm (Agility Digit v4, designed for logistics bin-picking) to 75 total DOF (1X NEO Beta, with 22 DOF per hand). This spread reflects a fundamental architectural divergence: logistics-oriented platforms optimize for payload and cycle time, whereas platforms targeting service or research applications invest in hand complexity.

Payload capacity spans 10–30 kg across the commercial tier, well above the requirements for most clinical manipulation tasks (stethoscope, ultrasound probe, catheter, laparoscopic instrument). The binding constraint is not force output but force *control*: clinical contact requires sub-Newton precision at the contact surface, a requirement that no platform has validated against patient-contact standards such as ISO/TS 15066. The single exception is Sanctuary Phoenix Gen 8, which specifies 5 mN tactile force sensitivity, the only commercial platform with a disclosed haptic sensing specification relevant to clinical touch discrimination.

Documented medical procedure performance exists for one platform only: the Unitree G1, in two independent studies achieving approximately 70% aggregate task success across seven clinical procedures under teleoperation [4], and demonstrating endoscope holding in a cadaveric surgical context [21]. Both results are preclinical prototypes. The cross-platform inference is sobering: the most dexterous commercial platforms (NEO Beta, Sanctuary Phoenix) have no clinical task evidence, while the only platform with clinical evidence (Unitree G1) was selected not for its dexterity but for its open SDK and research accessibility. Manipulation capability is not the bottleneck; validated manipulation *for clinical contact* is.

3.2 Locomotion and Clinical Environment Compatibility

The 2022–2024 generation of commercial humanoids completed a decisive architectural transition from hydraulic to electric actuation. All thirteen commercial platforms surveyed are fully electric, with brushless DC or high-torque servo actuators. This transition has three clinical implications: (1) reduced acoustic output enabling patient-proximate operation, (2) elimination of fluid-system leak risk in sterile environments, and (3) simplified field maintenance without hydraulic servicing. 1X NEO Beta specifies 22 dB operating noise (below the 30–35 dB threshold typical of hospital wards at night), making it the only platform with a disclosed acoustic specification relevant to patient care settings.

However, the clinical environment poses locomotion requirements that industrial deployments have not exercised. Hospital flooring transitions between vinyl, linoleum, ceramic tile, and threshold plates; patient room layouts require operation in spaces as narrow as 90 cm; and patient contact during transfer assistance imposes dynamic disturbance loads not present in logistics contexts. No commercial platform has published locomotion performance data in hospital environments, and no platform has been evaluated against IEC 62133 or IEC 60601 electrical safety standards in clinical settings.

Platform scale is a further structural constraint. The scope of this review sets a 150 cm lower bound for human-adult-care tasks; of the 13 commercial platforms, UBTECH Walker X (130–145 cm) and Unitree G1 (132 cm) fall at or below this threshold. The Unitree G1 is retained as the only platform with clinical evidence despite this limitation. Academic platforms designed for healthcare, including RHP Friends (168 cm, 54 kg, 30 DOF), demonstrate that human-scale form factors can be achieved with explicit clinical design intent [7].

3.3 LLM/VLA Integration and Task Architecture

Twelve of the thirteen commercial platforms surveyed include announced or deployed LLM/VLA integration [16]. Architectures span three patterns: (1) in-house VLA foundation models trained on proprietary demonstration data (Figure 02, Tesla Optimus); (2) integration of open third-party models such as Google Gemini Robotics (Boston Dynamics Atlas), open VLA frameworks (Unitree G1 with LeRobot and OpenVLA support), or Microsoft-partnered AI (Sanctuary Phoenix via Carbon); and (3) multimodal LLM integration for instruction following without explicit VLA policy training (UBTECH Walker X).

This architectural diversity obscures a uniform empirical gap: no VLA or LLM system deployed on any humanoid platform has been trained on, evaluated against, or validated for clinical task performance. The largest open VLA training corpus (Open X-Embodiment, approximately 970,000 demonstrations) consists entirely of household tabletop manipulation and pick-and-place tasks. Medical procedures require physically and procedurally distinct demonstration data (instrument grasps under sterile technique, force profiles during tissue contact, workflow sequencing under infection control constraints), none of which appear in any published training corpus.

The implication for clinical deployment is concrete: LLM/VLA integration as currently implemented provides natural-language task specification and general manipulation priors, but does not constitute a path to autonomous clinical task execution. The Unitree G1's teleoperation results [4, 21] are instructive precisely because they bypass VLA autonomy entirely: the cognitive and dexterous work is performed by a remote human operator, with the robot serving as a physical proxy. Universal and dexterous human-to-humanoid whole-body teleoperation has been demonstrated to enable complex bimanual tasks [32], consistent with the teleoperation-first architecture these clinical studies employ. This teleoperation-first architecture may represent the viable near-term clinical deployment model for all platforms, regardless of their onboard AI sophistication.

3.4 Platform Accessibility for Clinical Research

Research access to humanoid platforms is heavily asymmetric. Unitree Robotics is the only major commercial manufacturer to release and maintain a fully open hardware SDK, which directly explains why the Unitree G1 is the only commercial platform with peer-reviewed clinical task evidence. All other commercial platforms operate under proprietary software stacks with no published API access for external researchers. Academic platforms, including RHP Friends (30 DOF, purpose-built for nursing) and iCub3 (54 DOF, full-body tactile skin, 40+ units in research labs), provide broader research access but lack the scale and industrial backing needed for clinical deployment pathways.

The open-source software ecosystem relevant to humanoid clinical research is mature in general manipulation infrastructure but entirely absent in clinical-specific components. A systematic survey of prominent repositories yields four layers:

Foundation models and VLA frameworks. LeRobot [14] (HuggingFace; Apache 2.0) provides a hardware-agnostic imitation learning and VLA pipeline with direct Unitree G1 support. Physical Intelligence's openpi provides the π_0 [9] VLA model pre-trained on over 10,000 hours of real robot demonstrations, with best-reported results on ALOHA bimanual benchmarks [100]. OpenVLA [44] (7B parameters, 970,000 real-world demonstrations from Open X-Embodiment [67]; MIT license) and Diffusion Policy [19] complete the general-purpose manipulation toolkit. All are general-purpose; none include healthcare-specific training data, safety validation, or clinical task benchmarks.

Simulation environments. Genesis (Apache 2.0) provides unified differentiable physics simulation; Isaac Lab (BSD-3-Clause) offers GPU-accelerated reinforcement learning with native Unitree G1/H1 support. Neither provides medical environment models, clinical workflow scenarios, or validated patient-contact physics.

Unitree hardware ecosystem. The Unitree GitHub organization maintains nine publicly accessible repositories including unitree_rl_gym, unitree_sdk2, and xr_teleoperate for full-body XR teleoperation, the software component used in Atar et al. [4]. Low-cost leader-follower teleoperation hardware such as GELLO [96] further lowers the barrier to clinical procedure research.

The clinical infrastructure gap. Four categories of infrastructure required for clinical deployment are absent from the open-source ecosystem: (1) a validated safety layer enforcing patient-contact force limits at runtime; (2) clinical workflow integration connecting humanoid actions to hospital EHR and nursing systems; (3) an open clinical demonstration dataset (every major VLA training corpus consists of household and tabletop tasks); and (4) a validated medical simulation environment with clinical instrument physics and patient anatomy proxies. The transition from general-purpose manipulation to clinically safe, task-verified medical assistance is primarily a software and validation challenge that the open-source community has not yet begun to address.

Table 2 summarizes the most prominent open-source repositories in this ecosystem.

4 What Healthcare Needs from Humanoid Robots

Each clinical role imposes distinct requirements on a humanoid platform. The 14 qualifying papers reviewed here are evidence of how far current systems come toward meeting those requirements, not the subject of this section in themselves. This framing changes the question from "what has been done?" to "how close is the field to what clinical deployment actually demands?" The gap between those two positions is the central finding of this review.

4.1 Clinical Procedures and Surgical Assistance

Clinical procedure assistance demands sub-millimeter manipulation precision, controlled interaction forces below tissue damage thresholds (typically <1–2 N for soft tissue contact), sterility compliance (no bare actuator surface in the sterile field), and consistent task execution over procedure durations of 20–120 minutes. For teleoperated roles, latency below 100 ms round-trip is required to avoid operator disorientation during fine tool manipulation [5, 55]. For autonomous

Table 2. Open-Source Software Ecosystem for Humanoid and Embodied-AI Robotics (March 2026). Medical Relevance: **H** = high (direct hardware/algorithm support for clinical candidate platforms), **M** = medium (transferable general capability), **L** = low (infrastructure/simulation only). No repository provides clinical-specific safety validation or medical workflow integration.

Repository	**Org / Author**	**Purpose**	**License**	**Notes**	**Med. Rel.**
lerobot	HuggingFace	VLA/imitation learning pipeline	Apache 2.0	Supports Unitree G1; no clinical data	H
openpi	Physical Intelligence	π_0 VLA foundation model	Apache 2.0	10k+ hrs robot demos; bimanual ALOHA	H
openvla	Stanford/Berkeley/TRI	7B-param open VLA model	MIT	970k demos; generalist manipulation	H
act	tonyzhaozh	Action Chunking Transformers	MIT	Original ALOHA bimanual policy code	H
diffusion_policy	Columbia/MIT	Diffusion-based robot policy	MIT	Imitation learning / VLA pipeline	M
Genesis	Genesis-Embodied-AI	Differentiable physics simulation	Apache 2.0	No medical environment models	M
IsaacLab	isaac-sim (NVIDIA)	GPU-accelerated RL + sim	BSD-3	G1/H1 native support; no clinical env.	M
unitree_rl_gym	Unitree Robotics	RL locomotion for Unitree robots	BSD-3	G1/H1/B2 locomotion training	H
xr_teleoperate	Unitree Robotics	Full-body XR teleoperation	Apache 2.0	Used in Atar et al. 2025 study	H
unitree_sdk2	Unitree Robotics	Hardware SDK for G1/H1/B2	Apache 2.0	Low-level hardware interface	H
ManiSkill	HKUST	Embodied manipulation benchmark	Apache 2.0	No clinical task benchmarks	M
RoboManip	isri-robot	Bimanual manipulation dataset	MIT	Tabletop tasks only	M
Open Duck Mini	apirrone	Miniature bipedal robot (open HW)	MIT	Educational; not clinical scale	L

or semi-autonomous roles, real-time anatomical landmark detection and adaptive trajectory generation under tissue deformation are prerequisites.

Current evidence documents how far the field has advanced toward these requirements, and where the gaps remain. Three studies, all using the Unitree G1 under teleoperation, demonstrate feasibility across a range of procedures: Atar et al. [4] achieved approximately 70% task success across seven clinical procedures (auscultation, ultrasound, intravenous catheter insertion, and four others) with non-clinician teleoperators; Liang et al. [49] demonstrated laparoscopic instrument manipulation with inverse-kinematics remapping and enforced remote-center-of-motion constraint; and Cho et al. [21] performed cadaveric endoscopic surgery with a humanoid as first assistant. The robot's contribution in each case is physical presence and dexterity; clinical judgment and real-time adaptation are supplied by the remote operator. Tasks requiring high force precision (catheter insertion, finger-stick testing) showed the highest failure rates, attributable to actuator force resolution limits and insufficient tactile feedback.

The requirements gap is precise. Teleoperation architecture meets the near-term procedure assistance requirement, but the 70% task success rate is insufficient for unassisted clinical use. No study has measured forces in tissue contact against published safety thresholds. All evidence is pre-clinical (mannequins, cadavers, test fixtures); no patient data exist. Independent replication on any platform other than the Unitree G1 has not occurred. **TRL: 3–4.**

After the March 2026 search cutoff for this review, Zhang et al. [99] demonstrated a teleoperated rapid instrument exchange system for humanoid robots in MIS, exploiting the dual-arm configuration for natural surgical workflow. The system achieved stable instrument exchanges in constrained clinical environments, with novice operators converging rapidly after brief training. This work extends the surgical humanoid evidence sequence into instrument workflow management, a practical sub-problem that complements the procedure-execution demonstrations of Atar et al. [4] and Liang et al. [49].

4.2 Elderly and Nursing Care

Nursing care imposes a demanding combination of physical and social requirements. On the physical side: safe patient handling (transfer, repositioning, ambulation assist) requires payload capacity of 30–80 kg with compliant force control; gait stability on wet or uneven ward surfaces is required; and proximity to patients with fragile physiology demands contact-force limits more restrictive than industrial safety standards. On the social side: sustained interaction quality (naturalness, responsiveness, memory across sessions) matters for patient trust and care acceptance; eye-level engagement at standing height (150–170 cm) is preferred over wheeled alternatives for dignified interaction.

Against these requirements, the evidence reveals a clear bottleneck. The only bipedal humanoid designed explicitly for nursing is RHP Friends [7], which demonstrates autonomous-to-teleoperated handoff and nursing-specific motion primitives, but in controlled exhibition conditions, not patient-facing deployment. The SPRING project provides the only large-scale eldercare deployment data (60+ participants over 4 years), but on a wheeled ARI platform [2]; longer-term telepresence robot deployments in eldercare settings have similarly relied on wheeled systems [17]; participants' preference for upright humanoid form factor for eye-level interaction establishes a user acceptance hypothesis that bipedal systems have not yet verified with their own evidence. Imtiaz and Khan [36] surveyed 269 US nursing home administrators and found the primary deployment barriers are economic (cost, lack of effectiveness data), not technical; Mitzner et al. [59] similarly found that older adults' concerns center on capability and integration rather than technology rejection, a finding that reframes the investment priority: the field must produce health-economic evidence, not only engineering demonstrations. **TRL: 4 for demonstrated hardware; TRL 4–5 for wheeled platform deployment evidence (scope-constrained).**

4.3 Rehabilitation

Rehabilitation demands contact safety as a non-negotiable prerequisite: no patient-contact study on a full-size bipedal humanoid is ethically or regulatorily feasible without a validated compliance framework that bounds peak interaction forces. Beyond safety, repetition consistency across hundreds of movement cycles (neuromotor rehabilitation protocols typically require 300–500 repetitions per session), patient-adaptive trajectory modification in response to effort and fatigue, and integration with standardized outcome measures (Fugl-Meyer Assessment, 10-meter walk test) define the clinical requirements for this role. The evidence bar for robotic rehabilitation is well established: Lo et al. [53] demonstrated in a multi-site randomized trial that robot-assisted therapy for post-stroke upper-limb impairment can achieve outcomes comparable to standard care; the MIT-Manus neurorehabilitation robot [46] and ARMin exoskeleton RCT of Klamroth-Marganska et al. [45] similarly demonstrated TRL 7–8 performance for upper-limb rehabilitation. Robotic rehabilitation technology reviews confirm this trajectory [71]. These serve as the reference standard that bipedal humanoid rehabilitation systems must eventually match.

The requirements gap here is in two layers. The first is the safety precondition. Lu et al. [54] address it directly, demonstrating measurable peak force reduction during quasi-static upper-body contact under the GentleHumanoid spring compliance framework on the Unitree G1. This is the necessary engineering precondition for all patient-contact rehabilitation work that follows, but it has not been validated in dynamic motion or replicated on other platforms. The second layer is evidence of therapeutic efficacy. Nguyen et al. [65] establish a neuroscientific rationale: humanoid observation elicits sensorimotor EEG responses in healthy subjects consistent with action observation therapy (AOT) for stroke rehabilitation. Sobrepera et al. [86] ($n = 6$) show that patients preferred humanoid telepresence over video for neuromotor rehabilitation, attributing this to physical embodiment. The evidence is pre-clinical throughout: the

EEG study enrolled healthy subjects only; no study applies a validated clinical outcome measure; no study has involved patient contact with a current-generation full-size bipedal platform. **TRL: 3–4.**

4.4 Mental Health and Social Interaction

Mental health support demands naturalness in unscripted conversation, consistent interaction quality across sessions, and behavioral adaptation to individual patient affect and disclosure patterns. Trust is a functional requirement, not a preference: a patient who distrusts the system will not disclose, making the interaction therapeutically inert. For clinical deployment, outcomes must be measurable with validated instruments (PHQ-9, GAD-7, PANSS) in clinically diagnosed populations.

The evidence presents an inverse quality-specificity tradeoff. The best-designed studies are randomized: Robinson et al. [78] report a pilot RCT ($n = 230$) in which an autonomous humanoid delivered a mindful breathing intervention with positive wellbeing outcomes; Sayis and Gunes [80] find that only the humanoid robot condition produced statistically significant mood improvements and increased self-disclosure ($n = 42$). Neither study specifies the humanoid platform, which prevents extrapolation to full-size bipedal systems. The only full-size platform study (Ghosh et al. [29], $n = 14$) uses NAO (58 cm, out of scope for physical care roles). All studies use subclinical or university wellness populations; none involves a clinically diagnosed population under treatment. The embodiment advantage is consistent across designs, but it has been demonstrated on unspecified or out-of-scope platforms, in non-clinical settings, with non-patients. Yuan et al. [98] provide a transferable LLM personalization framework for dementia care on Pepper, and Lindsay et al. [50] demonstrate automated clinical planning in pediatric settings; both represent relevant methodological precedent for bipedal deployment. **TRL: 4–5 for social wellbeing applications in non-clinical settings; TRL 2–3 for bipedal humanoid deployment in clinical mental health contexts.**

4.5 Hospital Logistics

Hospital logistics requires reliable wayfinding in dynamic ward environments, payload capacity for transport tasks (medication carts, linen, supply bins up to 20–30 kg), and compatibility with existing hospital information systems for task scheduling. Crucially, it does not require bimanual dexterity, standing height, or natural language sophistication, which raises the threshold question: is the humanoid form factor the right choice for this role at all?

No qualifying papers were found for bipedal humanoid hospital logistics in the 2022–2026 period. Wheeled autonomous mobile robots already address this role with proven operational reliability; they provide no demonstrated functional disadvantage for logistics tasks, and their simpler regulatory profile (autonomous ground vehicle versus medical device) means faster deployment. The humanoid form factor offers no structural advantage for corridor navigation, supply transport, or medication delivery over wheeled alternatives. A study of cobot logistics in a 900-bed hospital system found that nurses report primarily workflow integration concerns rather than form-factor preferences [8]. Academic research attention in 2022–2026 was directed toward clinical procedure and HRI domains, reflecting this assessment implicitly. **TRL: 2 for bipedal humanoid hospital logistics; the role is better addressed by non-humanoid platforms at current maturity.**

4.6 Requirements Gap Analysis

Table 3 summarizes the requirements-versus-evidence position across all five domains.

Three structural patterns constrain the generalizability of the entire evidence base. **Platform monoculture:** the Unitree G1 is the experimental platform in 4 of 14 papers (all three procedure studies and the compliance rehabilitation

Table 3. Requirements gap by clinical domain (March 2026). TRL per ISO 16290:2013: 1 = basic principles; 2 = concept formulated; 3 = experimental PoC; 4 = lab validated; 5 = relevant-environment validated; 7 = operational prototype; 9 = proven. *ARI/SPRING: wheeled platform, not bipedal; scope caveat applies.

Domain	Key Papers	Platform	Study Type	N	TRL
Clinical procedures	Atar et al. [4], Liang et al. [49], Cho et al. [21]	Unitree G1	Lab; user study; cadaver	0 patients	3–4
Elderly / nursing	Benallegue et al. [7], Alameda-Pineda et al.* [2]	RHP Friends; ARI*	Exhibition demo; field deploy.	0 (demo); 60+*	4; 4–5*
Rehabilitation	Nguyen et al. [65], Sobrepera et al. [86], Lu et al. [54]	Humanoid; SRAT; Unitree G1	EEG pilot; case series; engineering	0 / 6 / 0	3–4
Mental health / HRI	Robinson et al. [78], Sayis and Gunes [80]	Unspecified humanoid	RCT; comparative	230; 42	4–5
Hospital logistics	*No qualifying papers; wheeled systems dominate (TRL 2)*				2

study). No other commercial platform appears in more than one qualifying study, because the Unitree G1's open SDK is currently the only commercial research access point; until equivalent access exists for higher-dexterity platforms (Sanctuary Phoenix Gen 8, 1X NEO Beta), cross-platform replication is structurally blocked. **Geographic concentration:** studies originate predominantly from the USA (8 papers), Japan (2), and the UK (2); no study from continental Europe (excluding the SPRING program), the Global South, or Southeast Asia appears in the qualifying set. Healthcare delivery models, regulatory frameworks, and workforce demographics differ substantially across these absent geographies. **Methodological ceiling:** no qualifying study constitutes a randomized controlled trial with a bipedal humanoid in a clinical population; no study applies a validated clinical outcome measure as its primary endpoint; no study exceeds a single session; no study includes health economic data. These gaps collectively define the minimum evidence package for any future regulatory submission: a prospective, multi-session, clinically powered study in a patient-contact context, with pre-registered primary endpoints and a costing analysis.

The domain-level TRL picture confirms that clinical procedures and rehabilitation share the same bottleneck (contact safety and force control), while elderly care and mental health share a different bottleneck: the bottleneck is not hardware but evidence. Both domains have transferred results from non-bipedal platforms (ARI/SPRING, unspecified humanoids) and non-clinical populations that must be replicated on full-size bipedal systems in patient populations before regulatory evidence packages can be assembled.

Figure 2 visualizes the distribution of the 14 qualifying papers by domain and year, with marker shape indicating study design type.

5 Challenges in Clinical Deployment

The TRL 3–4 state documented in Section 4 reflects three categories of barriers. Technical gaps include manipulation precision, contact safety during bipedal locomotion, and software validation against clinical performance standards. Institutional gaps include regulatory pathway ambiguity, liability framework absence, and actuarial non-coverage. A third category is circular: regulatory approval requires clinical evidence; clinical evidence requires Institutional Review Board (IRB)-sanctioned trials; IRB approval requires safety assurance; and safety assurance requires performance standards that do not yet exist.

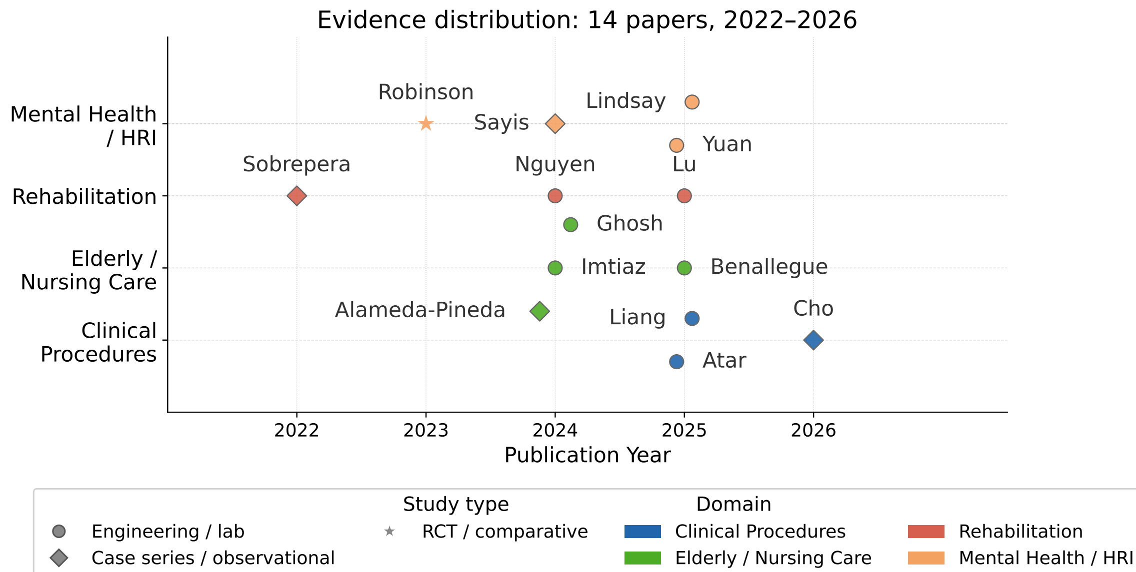

Fig. 2. Evidence distribution: 14 qualifying papers by clinical domain and publication year. Marker shape encodes study design (circle: engineering/lab; diamond: case series/observational; star: RCT/comparative); color encodes domain. Clinical procedure evidence is concentrated in 2025–2026. Rehabilitation and mental health evidence is more temporally distributed but limited by platform specificity. No papers qualify for hospital logistics.

5.1 Manipulation and Dexterity for Clinical Tools

Atar et al. [4] is the only study to systematically document manipulation performance gaps in a clinical task context. Atar et al. evaluated a Unitree G1 with Inspire Gen4 dexterous hands across seven clinical procedures, achieving approximately 70% aggregate procedural success with non-clinician teleoperators. The paper identifies two specific failure modes: force output insufficiency (the actuator configuration cannot reliably generate the controlled forces required for needle penetration or sustained grip against resistance), and sensor sensitivity limitations (the tactile feedback loop is insufficient for sub-Newton force resolution, preventing detection of the haptic signatures that guide needle procedures in clinical practice).

These findings generalize across the procedure literature. Liang et al. [49] demonstrate that laparoscopic surgery requires enforcing a remote-center-of-motion (RCM) constraint at the trocar site, an engineering workaround for a capability that commercial humanoid arms do not provide natively. Cho et al. [21] identify three engineering targets for endoscopic clinical translation: endoscope stability in confined anatomical cavities, force compliance during tissue contact, and positioning accuracy for reliable camera view.

Synthesizing across Atar et al., Liang et al., and Cho et al., clinical manipulation requires three capabilities that current commercial humanoid arms do not yet reliably provide: (1) force precision at sub-Newton resolution for needle and catheter procedures [4]; (2) tool-specific kinematic adaptation for laparoscopes, rigid endoscopes, and ultrasound probes; and (3) fingertip-level tactile sensing for grip force discrimination, instrument slip detection, and tissue compliance sensing. Operationally, progress requires force resolution at sub-Newton levels for needle insertion procedures [4], fingertip sensor density exceeding 1,000 sensing elements, and teleoperation round-trip latency at levels consistent with real-time force feedback requirements. Surgical teleoperation research establishes that network latency above

200 ms degrades task completion time and trajectory accuracy in dexterous procedures [55], providing the clinical benchmark for humanoid teleoperation system design. Foundational work on dexterous manipulation documents the multi-dimensional sensor requirements for reliable tool handling [66].

5.2 Safe Physical Contact and Bipedal Gait Dynamics

Progress in contact safety has been demonstrated. Lu et al. [54] introduce a spring-based compliance framework for the Unitree G1 that quantifiably reduces peak contact forces during quasi-static tasks including sit-to-stand assistance and upper-body physical contact. This approach builds on the series elastic actuator design [72], in which mechanical spring compliance decouples rapid force transients from the actuated structure. This is the most technically grounded progress toward patient-contact safety in the current literature. Hierarchical whole-body control architectures [83] provide the control-theoretic foundation for multi-priority task execution that safety-critical clinical interaction will require.

The design target for patient-contact force limits is the Power and Force Limiting (PFL) regime of ISO/TS 15066:2016 [41], which provides biomechanically derived thresholds for different body regions: for the face and skull, quasi-static contact force is limited to 65 N and pressure to 110 N/cm^2; for the trunk and thorax, 140 N and 210 N/cm^2. The GentleHumanoid framework demonstrates a Unitree G1 approaching these thresholds for quasi-static upper-body contact.

The critical gap, however, is structural. ISO/TS 15066 was designed for fixed-base collaborative robots (cobots whose base is anchored, whose motion is constrained to a defined workspace, and whose approach velocities are programmed). It does not model contact forces arising from bipedal locomotion. A bipedal humanoid navigating a clinical environment introduces four risk scenarios that ISO/TS 15066 does not bound: (1) balance corrections, in which whole-body torque adjustments generate rapid impulse forces that can contact nearby persons; (2) stumble recovery, in which momentum transfer creates swinging limbs and torso rotation at velocities above the standard's contact speed assumptions; (3) narrow-corridor navigation, in which swinging arms during gait will contact bed rails, IV poles, and patient limbs in spaces with less than one body-width of clearance; and (4) postural transitions, in which a full-size humanoid (35 kg or more) rising to or lowering from a working height near a supine patient generates dynamics that quasi-static force limits do not characterize.

None of the 14 papers reviewed addresses this category of risk, and neither ISO/TS 15066 [41], ISO 13482 [40], nor ISO 10218-1/2 [37, 38] was designed to model or bound gait-induced contact events. This is an open safety engineering problem. Resolution requires either new standards specific to bipedal humanoid robots in patient-contact environments, or architectural solutions (proximity-triggered gait interrupts, patient-detection zones constraining locomotion near occupied beds, whole-body collision prediction for gait trajectories) that have not yet been developed in the published literature.

5.3 Human-Robot Trust and Institutional Adoption

Trust and adoption operate at three distinct layers. The first layer is patient and clinician acceptance. Technology acceptance frameworks established for information systems [92] and their extensions for assistive social robots in elderly care [10, 34] provide the theoretical basis for understanding humanoid adoption barriers. Studies of robot acceptability in domestic and care settings further establish that transparency, predictability, and social role clarity determine acceptance [97]. Trust in human-robot interaction is a multidimensional construct influenced by robot performance, human factors, and environmental context [30, 31, 81]. The healthcare-specific question of whether humanoid form factors in clinical settings are perceived as appropriate has been characterized as a strategic and ethical question [70]. Robinson et al. [78] show 53% recruitment uptake for a humanoid wellbeing intervention in a general population (pilot RCT, n = 230). Sayis and Gunes [80] show the humanoid condition was the only condition producing measurable mood improvements in a therapeutic writing task (n = 42). Both results are bounded: the clinical populations central to humanoid deployment differ substantially from general and educational populations in these studies.

The second layer is co-design and workflow integration. Ghosh et al. [29] demonstrate that iterative co-design with care staff improves system usability, a methodological contribution that does not substitute for clinical evidence of safety and effectiveness.

The third and most institutionally consequential layer is the liability gap. No current healthcare liability framework (US tort law, UK clinical negligence, EU product liability) assigns accountability for harm caused by an autonomous robotic action in a care setting. This legal ambiguity deters adoption regardless of technical capability: hospital risk departments cannot quantify institutional exposure for systems operating in legally uncharted territory. Imtiaz and Khan [36] document this precisely: 269 nursing home administrators cite cost, reduction in human contact, and unproven clinical effectiveness as primary barriers.

5.4 Regulatory Pathway

The regulatory challenge is not hostility to this technology class; it is a jurisdictional gap that existing frameworks leave precisely where clinical humanoids sit.

ISO 13482:2014 [40] (Safety Requirements for Personal Care Robots) is the most directly applicable safety standard; it governs robots providing physical assistance in human environments. However, §1.4 of ISO 13482 explicitly excludes devices regulated as medical devices. A humanoid robot performing clinical tasks almost certainly meets the medical device definition under FDA 21 CFR §201(h) and EU MDR Article 2(1). This creates a hard gap: the system is likely too clinical for ISO 13482 but no ISO medical robotics standard covers general-purpose bipedal humanoid assistants.

In the United States, the De Novo classification pathway is the only viable regulatory entry route. No 510(k) predicate exists. A De Novo submission requires the applicant to propose Special Controls from scratch: accepted performance standards for fall prevention in patient-care environments, patient contact force limits for dynamic gait scenarios, teleoperation latency bounds for specific procedure types, and AI/ML software validation methodology consistent with FDA's AI/ML SaMD action plan [90] and December 2024 PCCP guidance [91]. None of these currently exist as accepted standards. This is not procedurally impossible: De Novo has established new device categories for surgical planning software and exoskeletal rehabilitation devices, but it represents a development and regulatory investment no humanoid manufacturer has yet made.

The structural problem is a circular dependency: approval requires clinical evidence; human-subject evidence requires IRB approval; IRB approval requires safety assurance; safety assurance requires performance standards; performance standards require regulatory guidance. The productive path through this circle is a staged evidence generation sequence (in vitro feasibility, cadaver study, IRB-approved human-subject feasibility trial) producing the data needed for a De Novo submission. Cho et al. [21] is the current frontier of that sequence.

In the EU, a humanoid clinical assistant would likely be classified as Class IIa under EU MDR Annex VIII, with autonomous surgical assistance escalating to Class IIb or III. EU AI Act Regulation 2024/1689 additionally classifies any AI-driven clinical humanoid as a high-risk AI system. High-risk AI rules apply from August 2026; AI embedded in regulated medical products transitions to August 2027. Any EU clinical humanoid market entry must satisfy both MDR and AI Act simultaneously, a dual compliance burden for which no completed template yet exists.

Table 4 compares the four principal frameworks across the key dimensions that determine market entry strategy.

Table 4. Regulatory Framework Comparison for Clinical Humanoid Robots (March 2026). PMS = post-market surveillance; PMCF = post-market clinical follow-up; SaMD = Software as a Medical Device; NB = Notified Body. No framework provides humanoid-specific guidance; all entries represent the authors' interpretation of existing rules applied to this novel device class.

Framework	Jurisdiction	Classification	Pathway	Timeline (est.)	Key Requirements / Gap
FDA 510(k)	US	Class I/II	Substantial equivalence to predicate	3–12 months	No predicate exists for clinical humanoid; inapplicable
FDA De Novo	US	Class II (new category)	Novel device; propose Special Controls	12–24 months	Must define fall-prevention, PFL, latency, AI/ML Special Controls from scratch
FDA PMA	US	Class III	Clinical trial evidence required	3–7 years	Overkill for initial classification; may apply for autonomous surgical
EU MDR Class IIa	EU	Medium risk	Notified Body QMS + clinical data	18–36 months	Requires clinical investigation under Art. 62; PMCF post-market
EU MDR Class IIb	EU	Higher risk	Notified Body + full clinical eval.	24–48 months	Autonomous surgical or patient-contact without operator oversight
EU AI Act (High Risk)	EU	High-risk AI	Conformity assessment; Art. 6	From Aug 2026	Parallel obligation to MDR; dual compliance burden for clinical humanoid
ISO 13482:2014	Global	Personal care robot safety	Voluntary standard	N/A	**Explicitly excludes medical devices**; no bipedal gait dynamic PFL spec
ISO/TS 15066:2016	Global	Collaborative robot HRC	Voluntary; used in EU MDR	N/A	Designed for fixed-base collaborative arms; no dynamic gait scenario coverage

5.5 Software Infrastructure and AI Validation

The leading VLA models, including openpi/π_0, OpenVLA [44], and LeRobot [14], represent the highest-capability general-purpose robot learning infrastructure publicly available. For general-purpose humanoid manipulation research, this infrastructure is genuinely strong.

What is absent is the layer between this general infrastructure and clinical deployment. As identified in Section 3.4, the four critical gaps with no current open-source solution are a clinical safety layer, clinical workflow integration, an open clinical demonstration dataset, and a validated medical simulation environment.

The AI validation gap extends to deployment readiness. VLA models achieve high success rates on standard benchmarks, but these share no tasks with the clinical procedure requirements documented in [4]. The distribution shift involves not only visual and geometric differences but also contact physics, tool compliance, and procedural sequencing requirements fundamentally different from household manipulation. AI in health and medicine more broadly faces similar validation challenges as these systems move toward clinical deployment [22, 75, 82, 89]. No humanoid software stack has addressed IEC 62443 Security Level 2 requirements either, the minimum threshold for networked systems in hospital environments.

Table 5 synthesizes the five principal technical and translational barriers identified in this section.

6 Discussion and Future Work

The evidence surveyed in Sections 4 and 5 shows a field that can demonstrate clinical relevance in laboratory settings but has not cleared any of the three thresholds required for operational deployment: accepted safety standards, clinical evidence in human subjects, and regulatory precedent.

Table 5. Technical and Translational Gap Summary (March 2026). Each row identifies a barrier from Section 5, its current state, the threshold required for TRL 5 advancement, which clinical domains it blocks, and the primary evidence or standard defining it. No single gap is insurmountable; they are sequentially dependent in the order listed.

Capability Gap	**Current State**	**Required for TRL 5**	**Blocking Domains**	**Key Evidence / Standard**
Force precision & tactile sensing	Insufficient sub-N resolution; no fingertip tactile array on commercial platforms	Sub-Newton force feedback; continuous fingertip sensing at clinical manipulation precision	Clinical procedures, Rehabilitation	[4]; [55]
Gait contact safety	ISO/TS 15066 quasi-static thresholds approached for upper body only [54]	Instrumented characterization of gait-induced force profiles; accepted standard for bipedal PFL	All physical-contact domains	ISO/TS 15066 [41]; ISO 13482 [40]
Tool-specific kinematics	No native RCM; no laparoscope-specific axis on commercial arms	RCM enforcement or software analog; tool-path control within trocar constraints	Clinical procedures	[49]; [21]
VLA clinical generalization	VLA benchmarks: household manipulation only; no clinical demonstration dataset exists	Open clinical demonstration dataset; VLA fine-tuned on clinical task protocols with validated performance	Clinical procedures, all autonomous roles	[67]; [100]
Clinical safety standard & regulatory predicate	No accepted standard for bipedal gait in patient-contact environments; no De Novo predicate	ISO scope revision; IRB feasibility study; FDA De Novo with proposed Special Controls	All domains	[40]; [90]; [91]

6.1 Technology Readiness Assessment

The TRL assignments in Table 3 provide the starting point. The aggregate finding (TRL 3–4 across all domains for bipedal humanoid platforms) is not a uniform statement; the TRL ceiling for each domain is governed by a specific domain-particular barrier.

In clinical procedures (current TRL 3–4), Atar et al. [4], Liang et al. [49], and Cho et al. [21] constitute the most technically substantive evidence in the review. The cadaveric study of Cho et al. is the current frontier of the evidence sequence: the step immediately preceding an IRB-approved human-subject feasibility trial. Advancing to TRL 5 requires two things simultaneously: a structured IRB-approved feasibility study in a specific narrow indication (telemedicine-assisted physical examination is the most defensible starting point, because fully teleoperated operation keeps a human in the decision loop at all times), and independent replication on a platform other than the Unitree G1.

For elderly and nursing care, the TRL gap between platform classes (4–5 for social robots; 3–4 for bipedal humanoids) is the central analytical finding for deployment planning. The SPRING project achieved TRL 4–5 through real-world deployment with 60+ older adult users over four years [2]. RHP Friends, the only purpose-built bipedal nursing humanoid, demonstrated patient transfer tasks at IREX 2023 but has not been evaluated in an actual care facility [7]. The path to TRL 5 for bipedal humanoids is not a question of platform capability but of deployment context: a structured pilot in an actual care facility, using SPRING-derived methodology on a bipedal platform, is the specific experiment that would advance this domain.

Rehabilitation (TRL 3–4) and mental health (TRL 4–5) share a structural problem: the strongest available evidence was generated on the wrong platforms. In rehabilitation, Nguyen et al. [65] establish neuroscientific rationale, Sobrepera et al. [86] establish clinical acceptability in a small case series, and Lu et al. [54] establish the contact-safety engineering foundation, but none of the three used a full-size modern bipedal humanoid with a clinical patient population. In mental health, Robinson et al. [78] ($n = 230$ RCT) and Sayis and Gunes [80] ($n = 42$) achieve methodologically strong results with consistent embodiment advantage findings, but neither specifies the humanoid model, and all participants are non-clinical wellness populations.

The advance required in both domains is the same experiment in different forms: an IRB-approved human-subject study using a specified full-size bipedal platform with a defined patient population. For rehabilitation, the target is a stroke trial with standardized outcome measures (Fugl-Meyer Assessment, 10-meter walk test). For mental health, the target is a feasibility trial with a clinically diagnosed population assessing validated scales. Both domains are one targeted study away from TRL 5. Both are currently blocked by the absence of that study, not by lack of underlying evidence.

Hospital logistics sits at TRL 2. No peer-reviewed evidence exists for humanoid-specific hospital logistics tasks. Wheeled autonomous mobile robots already address this domain with demonstrated operational reliability. Hospital logistics is not a priority for bipedal humanoid development at this stage.

Shortly after the literature search window closed, ZEALS and Omakase Robotics completed Japan's first hospital-based proof-of-concept deployment of a Unitree G1 at the University of Tsukuba Hospital (March 2026). The robot performed autonomous navigation, obstacle avoidance, and voice-guided delivery in the hospital lobby; the hospital director indicated intent to deploy within one to two years. This industry milestone, while not peer-reviewed, suggests that hospital logistics may advance faster than the published literature alone indicates.

Across all five domains, none has reached TRL 7 (system prototype demonstrated in operational environment). TRL 5 (the threshold of first human-subject validation in a controlled clinical setting) is the realistic five-year target for the three leading domains.

Figure 3 visualizes the current TRL state and TRL 5 target across all five clinical domains.

Figure 4 quantifies the gap between best current commercial performance and the scores required to reach TRL 5 for clinical deployment.

6.2 Near-Term Deployable Roles

Three roles have the strongest case for TRL 5 achievement within five years.

The first role is the telemedicine and telepresence clinical assistant. The teleoperated mode is the key risk-mitigation design choice: by keeping a human operator in the decision loop at all times, it avoids the regulatory classification and safety validation burdens associated with autonomous patient contact. The technical infrastructure exists: Atar et al. [4] demonstrate teleoperated clinical task execution across seven procedures, and Liang et al. [49] extend this to laparoscopic surgery. The primary application is specialist access extension: enabling a remote specialist to conduct a physical examination or assist in a procedure in a rural or low-resource setting. Realizing this role at TRL 5 requires a defined physical examination protocol, remote operator training standards, and IRB oversight.

The second role is the rehabilitation movement demonstrator. This role requires no direct patient contact: the humanoid demonstrates movements that the patient observes or follows. The bipedal gait safety gap identified in Section 5.2, the most universally blocking technical barrier, becomes irrelevant to this specific role, because the safety concern is dynamic contact force during locomotion near patients, not movement observation at distance.

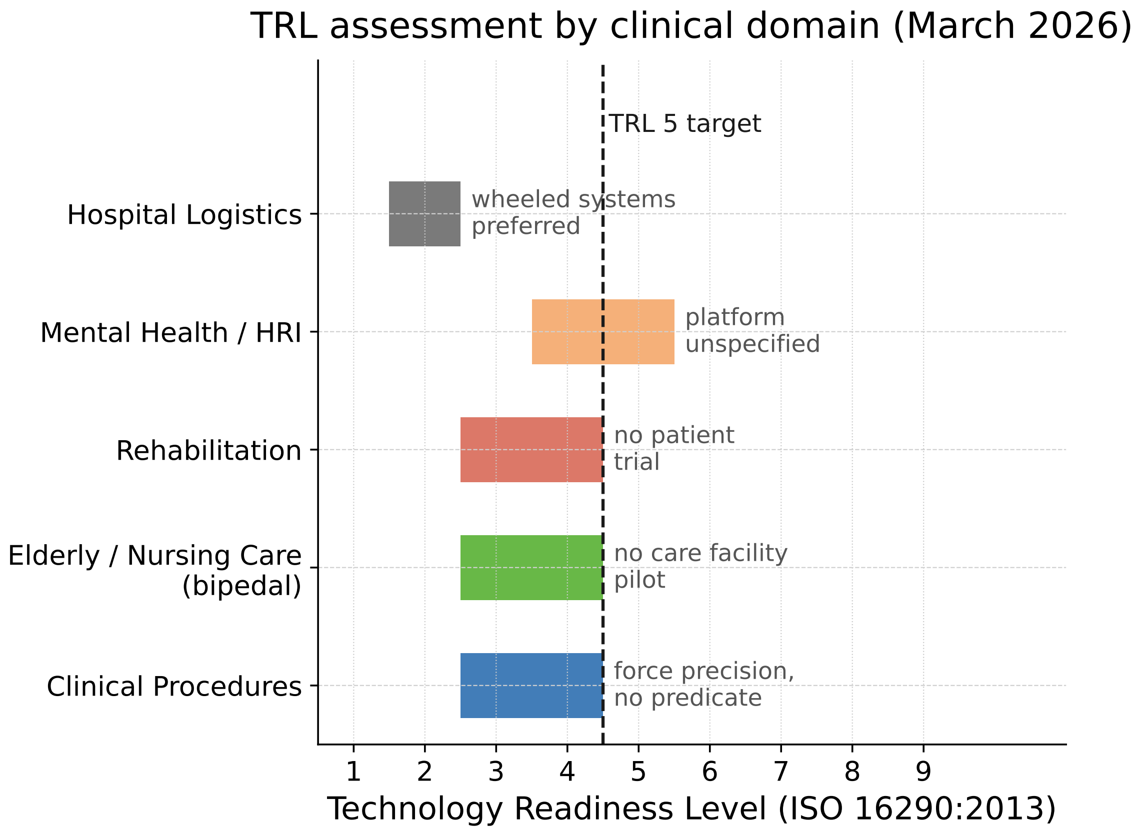

Fig. 3. Technology Readiness Level (TRL) assessment by clinical domain (March 2026). Current TRL range (filled bars) and TRL 5 target (dashed line) per ISO 16290:2013. No domain has reached TRL 7 (operational environment prototype); TRL 5 (relevant-environment validation) is the realistic five-year target for clinical procedures, elderly care (bipedal), and rehabilitation.

The neuroscientific rationale is grounded in Nguyen et al.'s mirror neuron evidence [65]; the clinical acceptability is established by Sobrepera et al.'s case series [86]. Advancing requires a registered feasibility trial adapting a NAO or Pepper protocol to a full-size bipedal humanoid demonstrator in stroke rehabilitation.

The third role is the structured eldercare companion and physical assistant. RHP Friends [7] is the only platform designed specifically for this role. The hybrid autonomous/teleoperated architecture (in which routine tasks run autonomously and non-routine tasks are handed to a remote operator) is the correct design for this deployment scenario. The SPRING project [2] provides the deployment methodology and user-acceptance evaluation framework. Realizing this role at TRL 5 requires an actual care facility pilot.

6.3 Research Priorities

Four priorities, ordered by their blocking impact on the entire field:

The first priority is clinical safety validation for bipedal gait in patient-contact environments. The four gait-induced contact risk scenarios documented in Section 5.2 are not bounded by any existing standard. No IRB will approve human-subject studies involving physical patient contact by a full-size bipedal humanoid without instrumented biomechanical

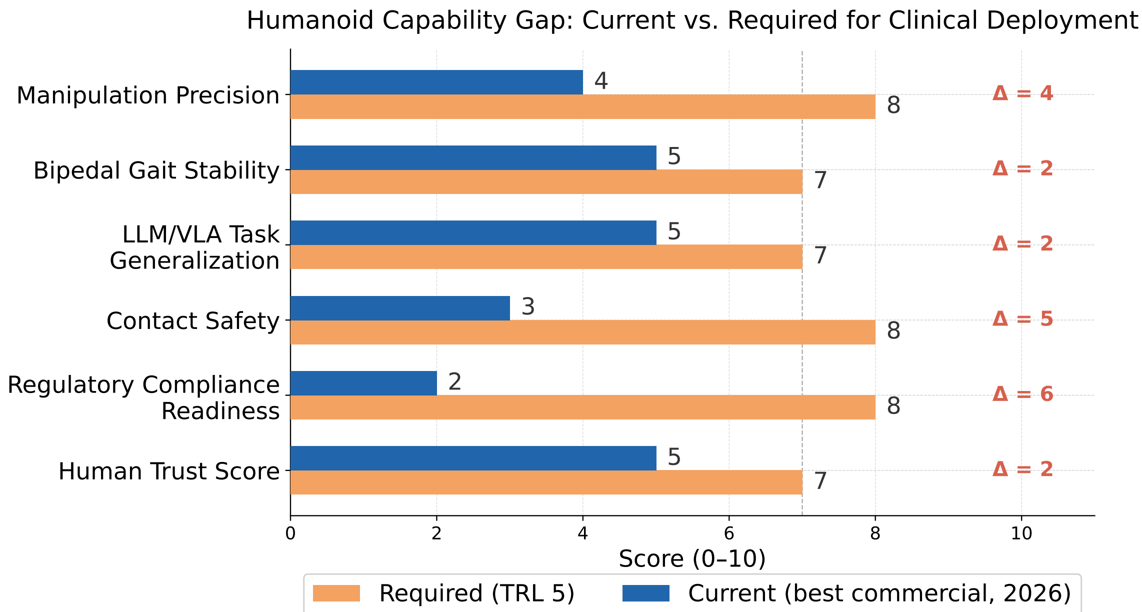

Fig. 4. Humanoid capability gap: current best-commercial scores (blue, 2026) versus required scores for TRL 5 clinical deployment (orange). Scale 0–10 represents engineering and evidence maturity. Regulatory Compliance Readiness and Contact Safety show the largest absolute gaps; Manipulation Precision is the dominant procedural engineering gap.

data on these force profiles and an accepted safety standard or architectural safeguard. This is the single most leverage-intensive investment the field can make: resolving it unblocks physical patient-contact studies in all three leading domains simultaneously.

The second priority is human-subject feasibility studies in leading applications. The cadaveric endoscopy study of Cho et al. [21] is the current clinical frontier. The specific next step is an IRB-approved, registered feasibility study in a narrow indication with an explicit safety monitor protocol. UC San Diego (Yip Laboratory) and Johns Hopkins hold the clinical infrastructure, cadaveric study capacity, and IRB precedent to conduct this work.

The third priority is an open clinical demonstration dataset for VLA training. No public dataset of humanoid robot demonstrations in clinical or clinical-simulated environments exists. Creating an open clinical demonstration dataset, even using simulators and clinical task protocols with mannequins, would unlock foundation model fine-tuning for clinical applications and enable the broader research community to advance clinical policy learning.

The fourth priority is platform diversification for evidence generalizability. The concentration of clinical procedure evidence on the Unitree G1 (Atar et al., Liang et al., Cho et al.) is a scientific limitation. Replication of clinical task evaluations on RHP Friends, TALOS, or other platforms would establish whether current findings are platform-specific or generalizable to the bipedal humanoid class.

6.4 Humanoid Clinical Deployment Readiness Framework

The TRL scale, originally defined by NASA for space systems [56] and later formalized as ISO 16290:2013 [39] for broader application [33], captures technology maturity along a single axis. Clinical deployment decisions require reasoning

Table 6. Humanoid Clinical Deployment Readiness (HCDR) framework applied to near-term deployable roles (March 2026). Binding constraint is the axis whose gap most delays deployment. TRL = Technology Readiness Level (ISO 16290:2013).

Role	Autonomy	Contact	Reg. Complexity	Binding Constraint
Telemedicine assistant	Teleoperated	Indirect (tools)	Class II / De Novo	Safety standard: bipedal gait
Rehabilitation demonstrator	Supervised	None	Class I–II	Evidence: patient trial
Eldercare companion / assist	Supervised-to-conditional	Direct	Class IIb–III	Both: gait safety + contact standard
Laparoscopic surgical assistant	Teleoperated	Indirect (instruments)	Class III	RCM precision + regulatory precedent

along at least three independent axes simultaneously: *autonomy level*, *patient-contact intensity*, and *regulatory complexity*. We introduce the Humanoid Clinical Deployment Readiness (HCDR) framework to make this three-dimensional space explicit. The framework is not a scoring system; it is a classification tool for locating specific role-platform combinations in the deployment readiness space and identifying which axis is the binding constraint.

Axis 1: Autonomy level. Roles range from fully teleoperated (human operator controls every action in real time) through supervised autonomy (the robot executes programmed sub-tasks under human oversight and with human abort authority) to conditional autonomy (robot operates autonomously under defined operational conditions with human escalation) and full autonomy (unsupervised operation). Current evidence supports only fully teleoperated roles. No qualifying study demonstrates supervised autonomy in a patient-contact task.

Axis 2: Patient-contact intensity. Contact ranges from no direct contact (observation, companionship, logistics) through indirect contact (physical interaction via tools or instruments, patient not in direct force chain) to direct physical contact (hands on patient, transfer assistance, examination). The binding regulatory and safety constraints scale sharply with contact intensity: indirect tool contact requires force control validation; direct physical contact requires validated safety standards that do not yet exist for bipedal platforms.

Axis 3: Regulatory complexity. Complexity is determined by device classification (Class I–III under FDA; Class I–III under EU MDR), presence of AI/ML as a decision-making component, and whether the application involves therapeutic intent (triggering clinical investigation requirements) or supportive logistics (lower classification pathway). Full autonomy with direct patient contact and therapeutic intent is the highest-complexity combination; teleoperation for physical examination assistance with physician oversight is among the lowest.

Table 6 maps the three near-term deployable roles against the HCDR framework and identifies the binding constraint on each path.

The framework reveals that the binding constraint is not uniform across roles. For the telemedicine assistant, the primary block is a missing safety standard for bipedal gait in patient-proximate environments, an engineering and standards-body problem, not a clinical evidence problem. For the rehabilitation demonstrator, the block is clinical evidence: the safety case is straightforward (no direct contact), but no powered clinical trial exists. For eldercare physical assistance, both gait safety and direct-contact force standards are missing simultaneously. The HCDR matrix operationalizes this distinction: investment in gait safety standards unblocks the telemedicine and eldercare roles; investment in a rehabilitation trial can proceed independently without waiting for standards resolution.

6.5 Practitioner Guidance

The evidence synthesized in this review has different implications for different practitioner audiences.

For clinical engineers and hospital technology leadership. The commercial humanoid platforms entering the market in 2024–2026 are designed for industrial labor, not healthcare. Procurement decisions made in 2026–2027 should distinguish between research-access platforms (Unitree G1: open SDK, lowest cost, only platform with published clinical task evidence, but physically limited) and future clinical platforms (RHP Friends: purpose-built but academic/prototype-only; Sanctuary Phoenix Gen 8: highest haptic sensing specification but closed ecosystem). No commercial platform is ready for unsupervised clinical use in any application. Any hospital-based evaluation program should structure the initial phase as an IRB- approved research protocol, not an operational pilot, to ensure safety monitoring, data collection, and liability clarity. Partnering with an academic medical center (not a robotics company) to co-design the evaluation protocol is the appropriate model.

For robotics engineers entering the healthcare domain. The most impactful contribution from the engineering community is not yet another manipulation algorithm or VLA architecture: those are not the bottleneck. Four specific infrastructure gaps documented in Section 3.4 represent engineering problems with no current open-source solutions: (1) a runtime safety layer enforcing ISO/TS 15066 contact force limits for bipedal dynamic scenarios; (2) clinical workflow integration middleware connecting humanoid action APIs to HL7 FHIR and hospital EHR systems; (3) a public clinical demonstration dataset for VLA fine-tuning on medical procedures; and (4) validated medical simulation environments with patient anatomy proxies and clinical instrument physics. Any of these would advance the field more than platform capability improvements at current TRL levels.

For clinical researchers and trial designers. Three study designs are currently feasible and would individually advance the field: (1) A registered, IRB-approved feasibility study of teleoperated humanoid physical examination assistance in a telemedicine-enabled rural clinic or outpatient setting, the lowest-complexity entry into patient-contact evidence. (2) A pilot RCT of humanoid-delivered action observation therapy for stroke rehabilitation, adapting the Sobrepera et al. protocol to a full-size bipedal platform and powered for a functional outcome measure (Fugl-Meyer Assessment). (3) A prospective multi-session care facility pilot adapting SPRING methodology to RHP Friends or an equivalent purpose-built platform, measuring care quality and staff burden outcomes. All three designs require a pre-specified safety monitoring protocol and an a priori definition of the study-stopping criteria for unexpected adverse events.

6.6 Regulatory Roadmap

Figure 5 illustrates the parallel regulatory tracks and their key decision points for both jurisdictions.

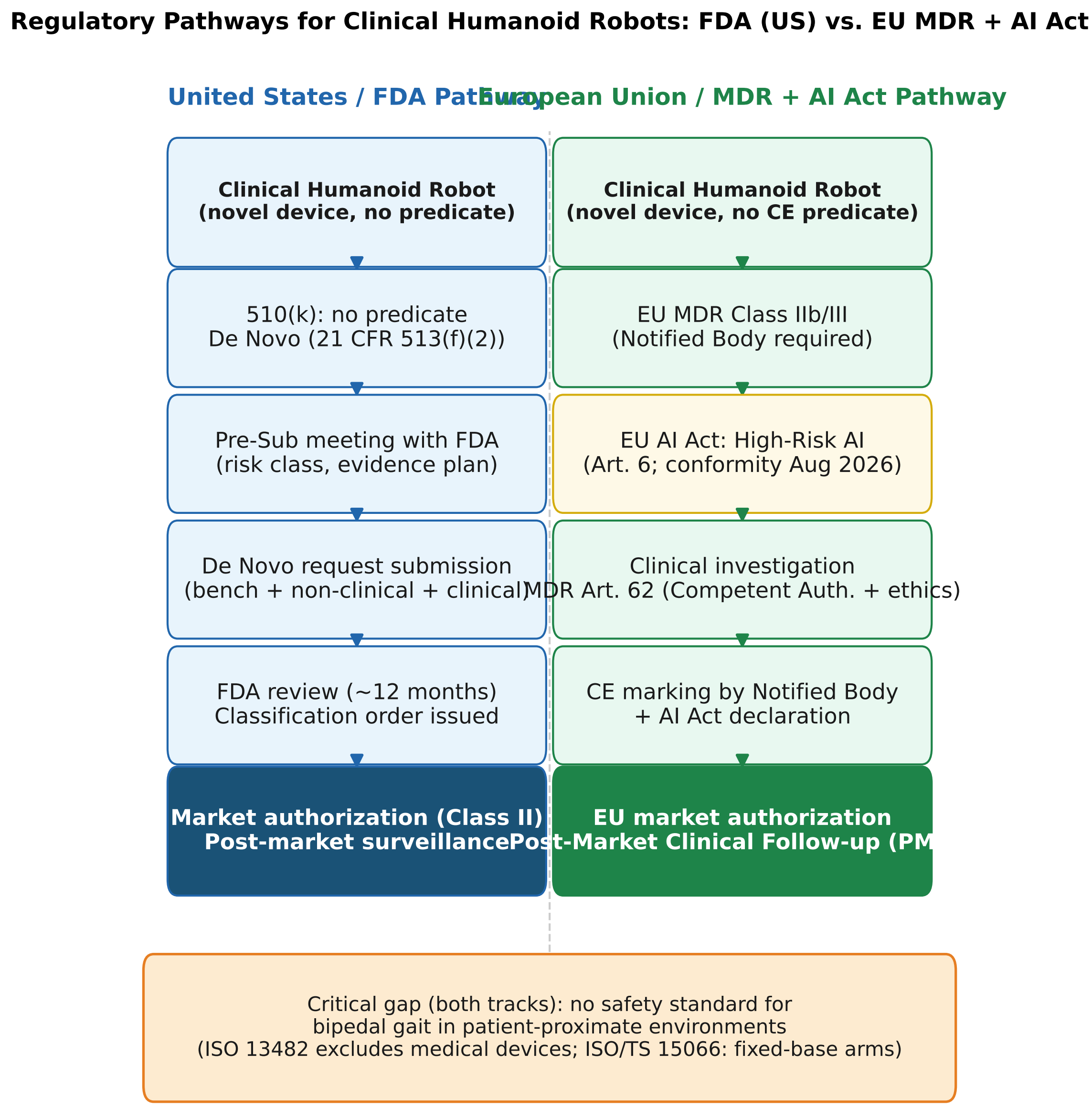

Fig. 5. Regulatory pathways for clinical humanoid robots: FDA De Novo (US, left) and EU MDR plus EU AI Act (EU, right). Both tracks require the same foundational evidence (safety standards for bipedal gait in patient-proximate environments) that does not yet exist (bottom banner). FDA De Novo is identified as the most plausible first-clearance pathway; EU MDR plus AI Act dual compliance represents the EU parallel track.

The path to the first regulatory clearance requires three sequential steps.

The first step is performance standards definition. A joint effort among humanoid manufacturers, academic medical centers (UCSD, JHU), and FDA's Center for Devices and Radiological Health (CDRH) is required to define the Special Controls that any De Novo submission must meet: fall prevention performance requirements, patient contact force limits for dynamic bipedal scenarios, teleoperation latency bounds, and AI/ML software validation methodology consistent with FDA's Good Machine Learning Practice principles and the December 2024 PCCP guidance.

The second step is an IRB feasibility study generating clinical evidence. A registered, IRB-approved feasibility study in a specific narrow indication generates the clinical evidence required for a De Novo submission. This study must be designed from the outset with regulatory submission in mind: pre-specified safety endpoints and a defined safety monitor protocol.

The third step is a De Novo submission that establishes a regulatory predicate. The De Novo application proposes a new product code for bipedal humanoid clinical assistants with Special Controls from Steps 1 and 2. First clearance establishes a predicate device, enabling subsequent 510(k) applications.

For the EU, the parallel path requires: an ISO/TC 299 scope revision of ISO 13482 to include clinical humanoids; EU MDR Class IIa conformity assessment; and EU AI Act compliance under high-risk AI obligations (effective August 2026 for new AI systems; August 2027 for AI embedded in regulated medical products). Regulatory precedents for managing layered framework compliance exist in the surgical robotics domain.

6.7 Ethical Considerations

Dignity of care. Physical care interactions require that the patient feels cared for, not processed. The SPRING project finding that participants cited eye-level conversational positioning as important for dignified interaction [2] provides direct empirical evidence that form factor affects perceived dignity of care. The uncanny valley phenomenon, the negative affective response elicited by human-like but not fully human entities [61], is particularly relevant for humanoid design in care settings, where distress rather than curiosity is the cost of a poor perceptual response. Care robotics designers must incorporate patient dignity explicitly as a design criterion alongside technical performance metrics.

Liability and informed consent. No current healthcare liability framework assigns accountability for harm caused by an autonomous or semi-autonomous robotic action in a care setting. The ethical dimensions of robot care for elderly individuals, including dignity, autonomy, and emotional manipulation risks, require explicit frameworks before deployment [84]. Before any human-subject deployment, consent processes for humanoid-assisted care must be defined. These require deliberate engagement between legal, ethical, and clinical stakeholders before the first human-subject study is launched.

Data privacy. A humanoid operating in a patient room integrates cameras, microphones, depth sensors, physiological monitors, and, in future deployments, integration with EHR systems, into a single mobile platform in one of the most privacy-sensitive environments imaginable. No data governance framework specific to clinical humanoid sensor data currently exists.

Workforce framing. The evidence does not support a displacement narrative. No clinical role exists that a humanoid currently performs reliably enough to replace a trained nurse or clinician. The evidence-supported framing is task offloading: routine, physically demanding, or cognitively repetitive tasks that do not require clinical judgment could be offloaded, freeing clinicians for judgment-requiring activities. The administrator survey of Imtiaz and Khan [36] shows that the perceived risk of reduced human contact is a primary institutional barrier; accurate communication of the task-offloading framing to institutional decision-makers is a parallel requirement, not a secondary one; neither the technical barriers nor the institutional perception barriers can be resolved in isolation.

7 Conclusion

As of March 2026, no bipedal humanoid robot holds FDA clearance or CE marking for any clinical application. No registered Phase I, II, or III clinical trial has enrolled patients for a study involving a full-size bipedal humanoid platform. The systematic search underlying this review identified 14 qualifying papers across 2022–2026, a literature sparse enough to read in its entirety, concentrated on a single commercial platform (Unitree G1) and, for the clinically most advanced domain, on a single research cluster (UCSD Yip Laboratory and Johns Hopkins). That concentration is not evidence of a field in failure; it is evidence of a field in its earliest phase of organized research. The infrastructure that would enable distributed clinical investigation (accepted safety standards for bipedal patient-contact environments, IRB-validated study protocols, and regulatory pathways with established Special Controls) does not yet exist, and its absence explains the concentration far better than any deficit of platform capability or clinical motivation.

The contribution of this review is to document that gap with specificity sufficient to act on it. Prior reviews of healthcare robotics, including the closest comparative work in the literature [23], either predate the LLM/VLA integration wave that is the enabling technical development of the current period, or treat humanoid and non-humanoid platforms interchangeably, obscuring the platform-specific barriers and advantages that define the clinical deployment problem. The synthesis here is the first to integrate the full 2022–2026 humanoid-specific literature with a commercial and academic platform capability assessment, a regulatory pathway analysis, and a per-domain TRL readiness evaluation grounded in clinical deployment criteria.

The field is not uniform in its readiness. Three roles have a credible path to first human-subject validation (TRL 5) within five years: a telemedicine and telepresence clinical assistant operating in fully teleoperated mode, which minimizes autonomous patient-contact risk and maps directly onto the established regulatory logic of physician-controlled surgical robotics; a rehabilitation movement demonstrator that exploits the mirror neuron mechanism established by Nguyen et al. [65] and the clinical acceptability demonstrated by Sobrepera et al. [86] without requiring any patient contact, rendering the bipedal gait safety gap irrelevant to this role; and a structured eldercare companion and physical assistant following the hybrid autonomous/teleoperated architecture demonstrated by RHP Friends [7] and the deployment methodology established by the SPRING project [2]. These are not speculative roles. They are identified by triangulating the strongest evidence in the literature, the most tractable technical barriers, and the lowest regulatory risk profile available to each application.

The HCDR framework introduced in Section 6.4 makes this role-differentiated picture precise. By classifying roles along three axes (autonomy level, patient-contact intensity, and regulatory complexity), the framework reveals that the binding constraint is not the same for every application. For the telemedicine assistant role, the block is a missing gait safety standard; for the rehabilitation demonstrator, it is clinical evidence; for eldercare physical assistance, both are required simultaneously. This decomposition has a practical consequence: investment in gait safety standards and investment in a rehabilitation feasibility trial are independent activities that can proceed in parallel. The field does not face a single bottleneck; it faces several distinct ones, each amenable to a focused, bounded intervention.

Two gaps are more consequential than the others identified in Section 5. The first is the absence of any accepted performance standard for bipedal humanoid robots in patient-proximate environments, specifically for the four categories of gait-induced contact events that ISO/TS 15066 was not designed to address. This gap does not require a breakthrough to close; it requires a biomechanical characterization study and the standards development process. But no IRB will approve physical patient-contact studies without it, and no regulatory agency can specify Special Controls without it. It is the single most globally blocking gap in the field. The second is the absence of registered human-subject feasibility studies. Cho et al. [21] advance the state of the art by completing a cadaveric surgical study under a defined IRB protocol, the step immediately preceding first-in-human research. UC San Diego and Johns Hopkins, which together hold all three papers in the clinical procedures domain, hold the necessary combination of clinical infrastructure, cadaveric study capacity, and IRB precedent for this next step. There is no institutional reason, beyond the gait safety gap, why this study cannot begin within the current funding cycle.

The question confronting the field is not whether humanoid robots can play a useful role in healthcare. The evidence in this review establishes that they can, not at the level of clinical deployment, but at the level of demonstrated laboratory capability in procedures and care tasks that have genuine clinical value. The question is whether the research community, device manufacturers, clinical institutions, and regulatory agencies will commit to the specific, sequential, methodical process required to convert that laboratory evidence into clinical evidence, and clinical evidence into regulatory precedent. That process (safety characterization, IRB feasibility study, De Novo submission, and predicate device establishment) is well understood in medical device development. Applying it to a genuinely novel platform class, with novel form-factor dynamics and novel AI system integration, requires institutional investment and coordination that has not yet materialized at scale. The evidence base is ready for the next step. The infrastructure to take it is the work of the next five years.

Acknowledgments

No external funding was received for this work.

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